



- (b) measurement**
- fixed 32-example probe, every 100 training steps
- **attention concentration**
fraction of 270 layer/query-head pairs above threshold
padding queries and query 0 excluded; sigmoid weights row-normalized
 - **value-norm ratio**
position 0 vs. all other valid positions, after value projection
3 KV-group norms pooled within a layer, then 30 layer ratios averaged
 - **residual-norm ratio**
position 0 vs. all other valid positions, at each block input
averaged across the 30 layers

- (c) conditions**
- **baseline**
softmax; random decoder
 - **g1gate**
softmax + zero-init gate (starts at 0.5)
 - **sigmoid**
unnormalized sigmoid attention
 - **textinit**
softmax; pretrained text decoder
 - **RF**
baseline arch.; fresh 1B-token stream
- g1gate and RF change more than one factor